

Week 4 Day 2

Write-up

Supervised Learning: Preprocessing, Models & Evaluation

Preprocessing

Before training the models, I prepared the data using a preprocessing pipeline to make sure the same steps were applied consistently during both training and testing.

Numeric features (age, education-num, capital-gain, capital-loss, and hours-per-week) were imputed using the median and then scaled with **StandardScaler**. **Median imputation** was chosen because it is less affected by outliers, especially since capital-gain and capital-loss are highly skewed.

For categorical features (workclass, marital-status, occupation, relationship, race, sex, and native-country), missing values were replaced with a new **"Missing"** category before applying **OneHotEncoder**. This allows the model to treat missing values as their own category instead of replacing them with the most common value. After comparing both approaches, using a "Missing" category produced slightly better results across all evaluation metrics, so it was selected for the final pipeline. OneHotEncoder was used because it converts categorical values into numerical features without introducing any false ordering between categories.

	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
most_frequent	0.8526	0.7386	0.5945	0.6588	0.7642	0.5361
Missing category	0.8553	0.7455	0.6001	0.6649	0.7678	0.5431

To simplify the dataset, education was removed because it contains the same information as education-num, and fnlwgt was also dropped since it was not expected to improve the prediction.

Model Comparison

So I trained **Logistic Regression** and **Decision Tree**. Both models used the same preprocessing pipeline and were evaluated on the hold-out test set using Accuracy, Precision, Recall, F1-score, ROC AUC, PR AUC, and Precision@K.

RESULTS:

	Accuracy	Precision	Recall	F1	ROC AUC	PR AUC
D1: Capital Gain Baseline	0.782	0.625	0.225	0.331	0.591	0.326
D2: Logistic Regression	0.855	0.745	0.600	0.665	0.905	0.765
D2: Decision Tree	0.823	0.640	0.599	0.619	0.775	0.512

Overall, **Logistic Regression** consistently outperformed the Decision Tree. It achieved higher Accuracy, Precision, F1-score, ROC AUC, and PR AUC while also producing fewer false positives. Since the project focuses on maximizing **Precision**, Logistic Regression better meets the business objective of identifying high-income individuals while reducing unnecessary marketing outreach.

The **Precision@K** results also supported this decision. Logistic Regression achieved **100% Precision** for the top 50 and top 100 predictions and maintained much higher Precision than the Decision Tree as K increased. This shows that the model is very reliable when selecting the highest-confidence individuals for targeted campaigns.

The confusion matrices showed that both models produced more **false negatives** than false positives. However, Logistic Regression generated significantly fewer false positives, making it a better fit for the chosen business objective.

The ROC and Precision-Recall curves also confirmed that Logistic Regression separates the two income classes much better than the Decision Tree.

Model Selection

Based on the evaluation results, I will continue working with **Logistic Regression** in the next stage. It performed better than the Decision Tree across almost every evaluation metric, including Accuracy, Precision, F1-score, ROC AUC, PR AUC, and Precision@K. It also produced fewer false positives, which is important because the main goal is to maximize Precision and avoid contacting people who do not actually earn more than \$50K.

Another advantage of Logistic Regression is that it is easier to interpret. The model's coefficients clearly show which features increase or decrease the likelihood of earning more than \$50K, making it easier to understand how the predictions are made.

For Day 3, I plan to experiment with preprocessing changes such as applying a log transformation to the skewed capital-gain. And I'll probably look into the categoricals and try target encoding to reduce dimensionality. I already dropped fnlwgt.